Khulna University of Engineering & Technology (KUET)
Department of Computer Science and Engineering

CSE 4102: Computer Graphics and Image Processing Laboratory

Final Project Report

3D Visualization of KUET Academic Building and Surrounding Campus

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1 Introduction

1.1 Project Aim

The aim was to construct an interactive 3D virtual environment representing the KUET academic core and surrounding campus infrastructure—including multi-story academic blocks, quadrangle residential halls, connected road networks, roundabout monuments, vehicles, and trees—that integrates as many core computer graphics principles as possible into a cohesive, interactive simulation . Every architectural and environmental asset is designed with structural intent: vehicles navigate campus roadways with dynamic wheel hierarchies, street lamps systematically illuminate transport corridors, and dynamic celestial orbits drive time-of-day light transitions and shadow projections across the scene .The project covers:

- **3D Modeling and Object Construction:** Creating 3D objects such as buildings, roads, bridges, cars, and buses using basic shapes like cubes, spheres, and cylinders. Combining these shapes and applying transformations to build complex objects
- **Symmetrical Building Design:** Creating symmetrical buildings, by using coordinate calculations to place objects on both sides without repeating the same design work
- **Lighting and Illumination:** Using different light sources, including the sun, moon, ambient light, and street lamps, to create realistic day and night scenes. Light intensity is adjusted based on distance and direction
- **Shading Techniques:** Implementing Flat, Gouraud, and Phong shading using GLSL shaders to control how light and colors appear on 3D objects
- **Shadow Mapping:** Creating realistic shadows using a two-pass rendering technique and a 2048×2048 shadow map. Percentage-Closer Filtering (PCF) is used to make shadow edges smoother
- **Textures and Materials:** Applying textures to surfaces such as roads and walls.
- **Interactive Camera and Navigation:** Allowing users to explore the 3D campus using an orbit camera with rotation and zoom controls, as well as a first-person camera for walking around the scene

1.2 Tools and Technologies

Language and build	C++ (compiled using standard `g++`)
Graphics API	OpenGL 3.3 Core Profile
Shading Language	GLSL 330 (`default.vert`, `default.frag`)
Libraries	GLFW (window and input management), GLAD (OpenGL function loading), GLM (mathematical operations)
GUI	Dear ImGui for interactive controls and real-time scene settings
Texture Management	Custom `Texture` class and `stb_image.h` for loading and applying textures
Window Resolution	1280×960 pixels
Rendering Features	Flat, Gouraud, and Phong shading, multiple light sources, and shadow mapping



(a) First-Person View



(b) Orbit Camera

Figure 1: Comparison of First-Person and Orbit Camera perspectives.

1.3 Coordinate System and Scene Dimensions

Coordinate System	Right-handed 3D Cartesian coordinate system
World Origin	$(0, 0, 0)$
Ground Plane	150×150 units

1.4 Camera System

Orbit Camera	Adjustable rotation and zoom, with a default radius of 40 units
First-Person Camera	Starts at $(3.5, 0.7, 44.0)$ and supports interactive movement
Perspective Projection	45° field of view, with near and far clipping planes at 0.1 and 150.0 units

1.5 Lighting and Shadows

Directional Light	Represents sunlight, positioned at $(40, 30, 15)$
Point Light	Positioned at $(-10, 16, 2)$
Street Lighting	26 spotlight sources for nighttime illumination
Shadow Mapping	Uses a 2048×2048 depth map for shadow generation
Day/Night Modes	Adjustable lighting and ambient illumination for different scene conditions

2 Primitive Geometry amp; Transformation Mechanics

2.1 Base Geometry Primitives

All 3D objects in the scene are built using four basic shapes: cubes, spheres, cylinders, and half-cylinders. These shapes are defined around the origin $(0, 0, 0)$ and scaled, rotated, and moved to create buildings, roads, vehicles, and other campus objects.

2.1.1 Unit Cube

- **Size:** Extends from -0.5 to $+0.5$ along the X, Y, and Z axes.
- **Geometry:** Consists of 6 faces divided into 12 triangles, with 36 vertices in total.
- **Vertex Data:** Each vertex contains its position, surface normal, and 2D texture coordinates.

2.1.2 Unit Sphere

- **Size:** Has a radius of 1.0 and is centered at the origin.
- **Geometry:** Generated using a UV sphere with 16 stacks and 24 sectors.
- **Vertex Position:** Each vertex is calculated using latitude and longitude angles:

$$x = \cos(\phi)\cos(\theta), \quad y = \sin(\phi), \quad z = \cos(\phi)\sin(\theta)$$

- **Surface Normals:** Since the sphere has a radius of 1.0, each vertex position can also be used as its unit surface normal.

2.1.3 Cylinder

- **Size:** Extends from -0.5 to $+0.5$ along the local X-axis, with a circular cross-section of radius 1.0.
- **Geometry:** Uses 24 segments to create the curved surface and two circular end caps.
- **Surface Normals:** The curved surface uses outward-facing radial normals, while the two end caps use normals pointing along the positive and negative X directions.

2.1.4 Half-Cylinder

- **Shape:** A half-circular shape with a radius of 1.0, extending vertically from $Y = -0.5$ to $Y = +0.5$.
- **Geometry:** Generated using 20 angular segments.
- **Components:** Includes the curved surface, top and bottom half-circle caps, and a flat rectangular back face.

2.2 Transformation Mechanics & Matrix Pipeline

World-space objects are instantiated by applying homogeneous 4×4 transformation matrices to unit primitives .

2.2.1 Standard Transformation Pipeline

To place a unit primitive with scaling $\mathbf{S} = (s_x, s_y, s_z)$, Euler rotation $\mathbf{R_a}(\alpha)$ about axis $\mathbf{a}$, and world translation $\mathbf{T} = (p_x, p_y, p_z)$, matrices are multiplied in strict right-to-left order :

$$\mathbf{M} = \mathbf{T}(p_x, p_y, p_z) \cdot \mathbf{R_a}(\alpha) \cdot \mathbf{S}(s_x, s_y, s_z)$$

In 4×4 homogeneous matrix representation :

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha & 0 \\ 0 & \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{R}_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.2.2 Normal Vector Transformation

Surface normals are important for calculating lighting and shading. When an object is scaled differently along different axes, its normals must be transformed correctly to remain perpendicular to the surface.

The vertex shader uses the inverse transpose of the model matrix:

$$N_{\text{world}} = \text{mat3}\left((M^{-1})^T\right) \cdot \text{aNormal}$$

The resulting normal is normalized before being used in lighting calculations.

2.2.3 Hierarchical Modeling

Hierarchical modeling allows complex objects to be built from smaller parts. Each child object uses its parent's transformation as the starting point for its own transformation.

For example, a vehicle consists of a body, windshield, and wheels. When the vehicle moves or rotates, all its parts move with it. The transformation of a vehicle is:

$$M_{\text{vehicle}} = T \cdot R_y$$

The body, windshield, and wheels then apply their own local transformations to this parent matrix.

Car and Bus Modeling

- **Vehicle Body:** Created by scaling and positioning a cube.
- **Windshield:** Created using a scaled cube with an additional rotation.
- **Wheels:** Positioned relative to the vehicle and rotated to simulate rolling motion.

The wheel rotation is updated as the vehicle moves, making the animation more realistic.

Street Light Modeling Street lights are built by combining several basic shapes:

- **Base:** Stacked cubes form the bottom section.
- **Vertical Pole:** A cylinder is rotated to align it with the Y-axis.
- **Cross-Arm:** Extends horizontally from the pole.
- **Lamp Heads:** Spheres and other shapes create the lamp covers and light bulbs.

The components are positioned relative to one another to form a complete street light.

2.3 Symmetrical Building Design

The Residential Hall BRH is created using the reference of the hall FFH/DRH. Instead of rotating every component separately, the building coordinates are calculated to create a symmetrical layout around the center point $(27, 23)$ in the XZ-plane.

The reflected coordinates are calculated as:

$$x' = 2x_c - x = 54 - x$$

$$z' = 2z_c - z = 46 - z$$

The Y-coordinate remains unchanged.

This method makes it easier to create matching building layouts, including walls, windows, doors, stairs, and other architectural details, while reducing repeated transformation calculations.

3 Scene Layout: Top, Front, Side and Isometric Views

World coordinates use X to the right, Y up and Z towards the main gate (south). The main road runs along Z at $X = 0$–7; the academic blocks stand east of it at $X = 15$, and the halls stand west of it. The ground is a $150 \times 0.1 \times 150$ slab. The views in this section were produced by an orthographic/isometric projection of the same coordinates and dimensions used in `main.cpp`, so they accurately show the layout, relative sizes and colours; they are schematic (flat-coloured) renderings and are not screenshots of the OpenGL window.

3.1 Top View (Plan)

From above, the main road (7 units wide) is the spine of the scene. The four academic blocks (8×4 each) are stacked in a row along Z on the east side, joined by sky-bridge corridors. The brick roundabout ring sits south of the blocks at $(15.2, 18)$. West of the main road lie the courtyard-shaped halls. Trees line both sides of the main road and the back road; twin-head lamp posts stand at regular intervals.

3.2 Front View

In the front elevation the tallest structures are the five-storey academic blocks (about 6.8 units) and the AEH/SSH halls (5.83 units). The FFH, DRH, LHS and BRH halls are 3.75 units high. Lamp posts reach 4.85 units, trees 4.8 units, the bus about 2.8 units and the car about 1.15 units.

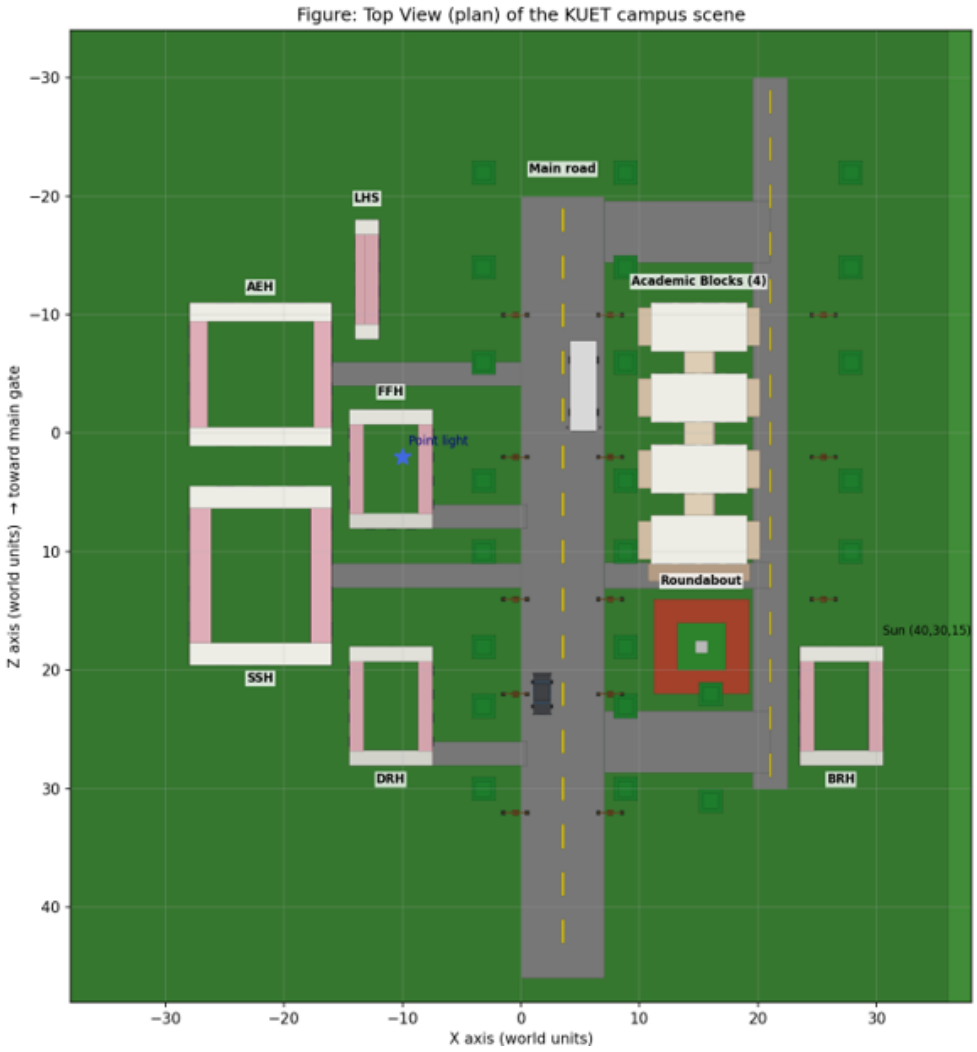


Figure 2: Top view of the campus: roads, halls (AEH, SSH, FFH, DRH, LHS, BRH), academic blocks, roundabout, trees, street lamps, vehicles and the position of the point light and sun.

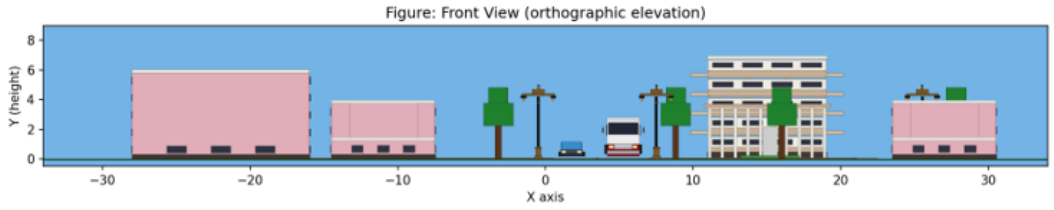


Figure 3: Front elevation (looking from $+Z$ towards $-Z$).

3.3 Side View

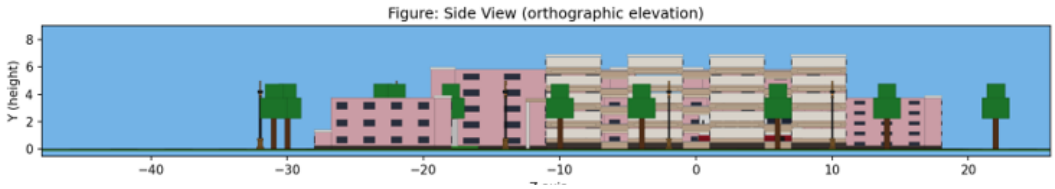


Figure 4: Side elevation view of the campus layout.

4 Summary of all Objects

Summary of every object in the scene, its location and its main parameters are given in the table below; each object is then described in detail with its front, side, top and isometric views, its shading and its texture.

Table 1: Summary of Scene Objects, Locations, and Parameters

Object	**Position** (X, Z)	**Overall size**	**Count / notes**
Ground	$(0, 0)$	$150 \times 0.1 \times 150$	1 slab, green $(0.25, 0.55, 0.22)$
Sun	$(40, 30, 15)$	Sphere, scale 6	Emissive; yellow by day, grey-blue at night
Academic blocks	$X = 15$; $Z = -9, -3, 3, 9$	$8 \times 6.5 \times 4$ each	4 blocks, 5 floors each
Sky-bridge corridors	$X = 15$; between blocks	2.6 wide, 2.2 long	3 corridors $\times$ 5 levels + roof slab
Front portico	$X = 15$, $Z =$ 11–12.5	8.6×3.6 high $\times 1.5$	1 canopy, 10 pillars
AEH hall	$(-22, -5)$	$12 \times 5.83 \times 12$, wall 1.5	5-storey courtyard building
SSH hall	$(-22, 12)$	$12 \times 5.83 \times 15$, wall 1.8	5-storey courtyard building
FFH hall	$(-11, 3)$	$7 \times 3.75 \times 10$, wall 1.2	3-storey, 1-storey front wing
DRH hall	$(-11, 23)$	$7 \times 3.75 \times 10$, wall 1.2	same model as FFH
LHS hall	$(-13, -13)$	$2 \times 3.75 \times 10$, wall 1.2	narrow 5-level block
BRH hall	$(27, 23)$	$7 \times 3.75 \times 10$, wall 1.2	FFH layout rotated 180°
Main road	$(3.5, 13)$	$7 \times 0.02 \times 66$	Textured, centre dashes every 4 units
Back road	$(21, 0)$	$3 \times 0.02 \times 60$	Textured, centre dashes

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Table 1 continued from previous page

Object	**Position** (X, Z)	**Overall size**	**Count / notes**
Link roads (8)	various	2–5.2 wide	Connect halls and blocks to the main road
Roundabout	$(15.2, 18)$	8×8 ring, island 4×4, monument $1\times2\times1$	Brick texture
Trees	$X = -3.2, 8.8, 27.8$ and $(16, 22)$, $(16, 31)$	$\approx 2 \times 4.8 \times 2$	23 trees
Street lamps	$X = -0.5, 7.5$ ($Z = -10, 2, 14, 22, 32$) and $X = 25.5$ ($Z = -10, 2, 14$)	4.85 high	13 posts, 26 lamp heads
Bus	Lane $X = 5.25/1.75$	$2.2 \times 2.78 \times 7.6$	Animated, 4 rotating wheels
Car	Opposite lane	$1.5 \times 1.15 \times 3.4$	Animated, 4 rotating wheels

5 Ground Plane, Roads & Environment Setup

5.1 Ground and Terrain Modeling

The campus ground is created using a large, flat cube placed at the world origin $(0, 0, 0)$.

- **Shape and Size:** A unit cube is scaled to $150 \times 0.1 \times 150$ units to create the ground surface.
- **Height:** The ground is centered at $Y = 0$, so its top surface is at $Y = 0.05$. This provides a reference level for buildings, roads, and trees.

5.2 Road and Transport Network

The campus road network is created using scaled cube primitives. It includes the main roads, connecting roads, residential hall access roads, and yellow center markings.

Road surfaces are placed slightly above the ground to reduce z-fighting, which occurs when two surfaces are at nearly the same depth.

5.2.1 Main Roads

Main Central Road

- **Position:** $(3.5, 0.05, 13.0)$
- **Size:** $7.0 \times 0.02 \times 66.0$ units.
- **Coverage:** Extends along the North–South direction, from $Z = -20$ to $Z = 46$.

- **Material:** Used texture.

Eastern Back Road

- **Position:** $(21.0, 0.05, 0.0)$
- **Size:** $3.0 \times 0.02 \times 60.0$ units.
- **Coverage:** Extends from $Z = -30$ to $Z = 30$.

Road textures use UV coordinates to control how frequently the texture repeats across the road surface.

5.3 Roundabout and Central Monument

The Durbar Bangla area is created at the road intersection near the center of the campus. It consists of three main parts: an outer road surface, a raised grass island, and a central monument.

5.4 Tree Modeling

Trees are created by combining simple geometric shapes. Each tree consists of a trunk and two layers of green foliage.

- Trunk : **Position:** $(x, 1.25, z)$ **Size:** $0.4 \times 2.5 \times 0.4$ units.
- Lower Foliage : **Position:** $(x, 3.0, z)$ **Size:** $2.0 \times 1.5 \times 2.0$ units.
- Upper Foliage : **Position:** $(x, 4.2, z)$ **Size:** $1.3 \times 1.2 \times 1.3$ units.

The two foliage layers are stacked above the trunk to create a fuller tree shape.

5.4.1 Tree Placement

Trees are placed along roads and around the roundabout to make the campus look greener and more realistic.

A total of 23 trees are specified in these four groups. Their placement helps frame the roads, enhance the landscape, and improve the overall appearance of the 3D campus scene.

6 Architectural Modeling of Academic Building Blocks

6.1 Overview & Modular Layout Concept

The central Academic Building is designed as a modular 4-block connected complex aligned vertically along the East axis at $X = 15.0$. Rather than rendering a single monolithic structure, the architecture is partitioned into four distinct 5-story building blocks situated along the North–South axis at $Z \in \{-9.0, -3.0, 3.0, 9.0\}$.

These four identical blocks are systematically integrated into a unified academic complex through two key architectural mechanisms:

1. **Multi-Story Skybridge Corridors :** Three elevated 5-story walkway corridors span the 2.0-unit gaps between adjacent blocks ($Z = -6.0, 0.0, +6.0$).

2. **Grand Entrance:** Attached to the front face of the southern block ($Z = 9.0$), a two-story overhang canopy supported by a 10-pillar colonnade extends the front facade to $Z = 12.35$, serving as the primary entrance to the academic complex.

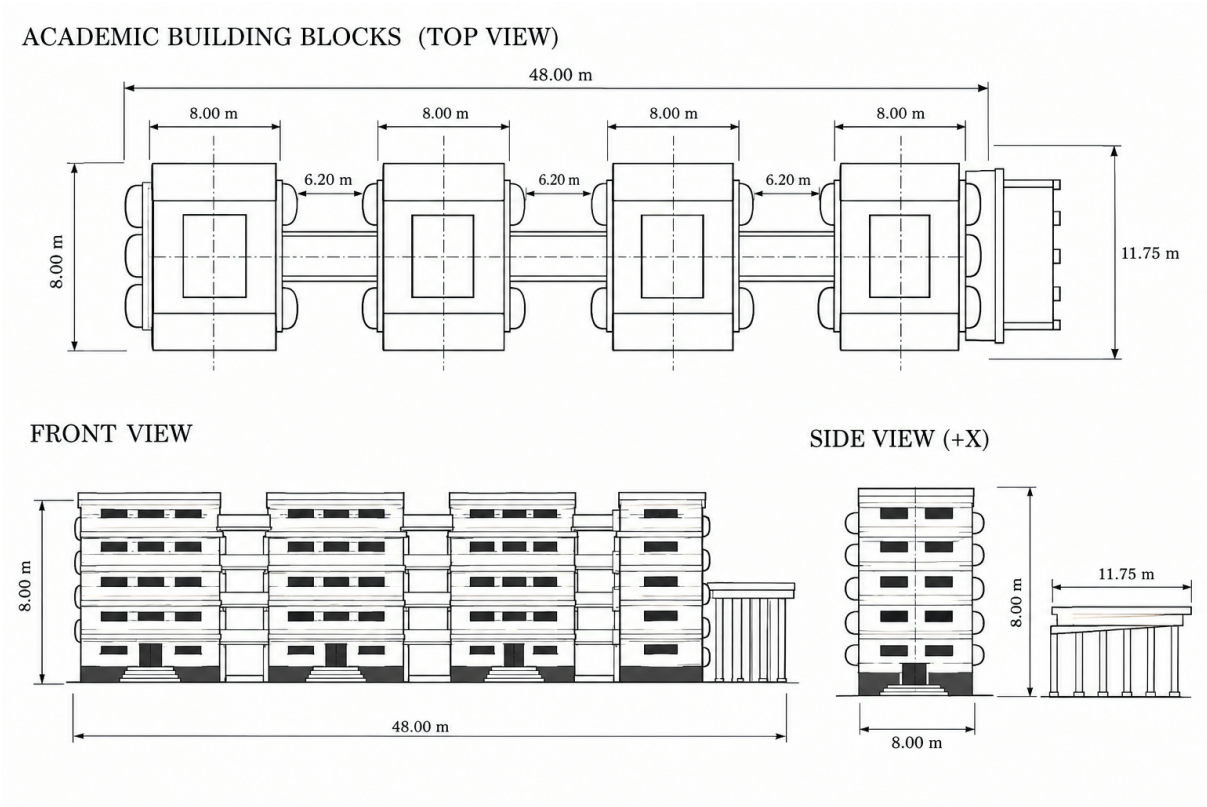


Figure 5: Modular Layout Concept of the 4-Block Academic Building

6.2 Primitive Geometries Required

The academic building complex is built entirely from two primitive mesh types:

- **Unit Cube Primitive :** Used for all rectangular structural members, including the foundational plinth base, wall facade bands, recessed window glass panes, entrance doors, stair treads, skybridge floor slabs, parapet walls, handrails, and colonnade pillars .

- **SemiOval / Half-Cylinder Primitive :** Used exclusively for the side oval balconies projecting outward from the $+X$ (East) and $-X$ (West) exterior walls across upper floors. It provides a curved architectural silhouette that breaks up the boxy rectangular walls.

6.3 Construction of a Single Building Block

Each individual block occupies a footprint of 8.0×4.0 world units and rises to a height of $H = 6.5$ units across 5 stories (1.3 units per story). It is constructed through the following sub-components:

1. **Dark Base Plinth:** A dark gray/brown base slab scaled to $\mathbf{S}(8.05, 0.50, 4.05)$ and translated to $(x_{\text{offset}}, 0.25, z_{\text{offset}})$, slightly wider than the building to form a grounded footing.

2. **Two-Tone Facade Layering (5 Floors):** For each floor level $f \in \{0, 1, 2, 3, 4\}$, the floor height is evaluated as $Y_{\text{floor}} = 0.8 + 1.3 \cdot f$:

   - **Lower Tan Wall Band:** Scaled to $\mathbf{S}(8.0, 0.50, 4.0)$ at $Y = Y_{\text{floor}} - 0.25$.
   - **Upper Cream Wall Band:** Scaled to $\mathbf{S}(8.0, 0.70, 4.0)$ at $Y = Y_{\text{floor}} + 0.35$.

3. **Horizontal Ribbon Windows:** Four dark glass window panes are recessed along the front $+Z$ face ($Z = z_{\text{offset}} + 2.01$) and back $-Z$ face ($Z = z_{\text{offset}} - 2.01$) per floor. The window X-positions are spaced parametrically at $X = x_{\text{offset}} + \text{win} \cdot 1.0$ for $\text{win} \in \{-3, -1, +1, +3\}$ with scale $\mathbf{S}(1.4, 0.35, 0.04)$.

4. **Side Oval Balconies:** Four semi-oval balconies are attached to both the $+X$ and $-X$ sides across floors 1 to 4 at heights $Y = i \cdot 1.3 + 0.5$ ($i \in \{1, 2, 3, 4\}$). The $+X$ balconies are rendered with zero rotation, while the $-X$ balconies apply a 180° Y-axis rotation matrix $\mathbf{R}_Y(180°)$ to orient the curved wall outward.

5. **Main Center Doors & Staircases:** Central entrance doors scaled to $\mathbf{S}(1.2, 1.5, 0.08)$ are placed on the ground floor at both $+Z$ and $-Z$ sides, flanked by 4-step concrete staircases.

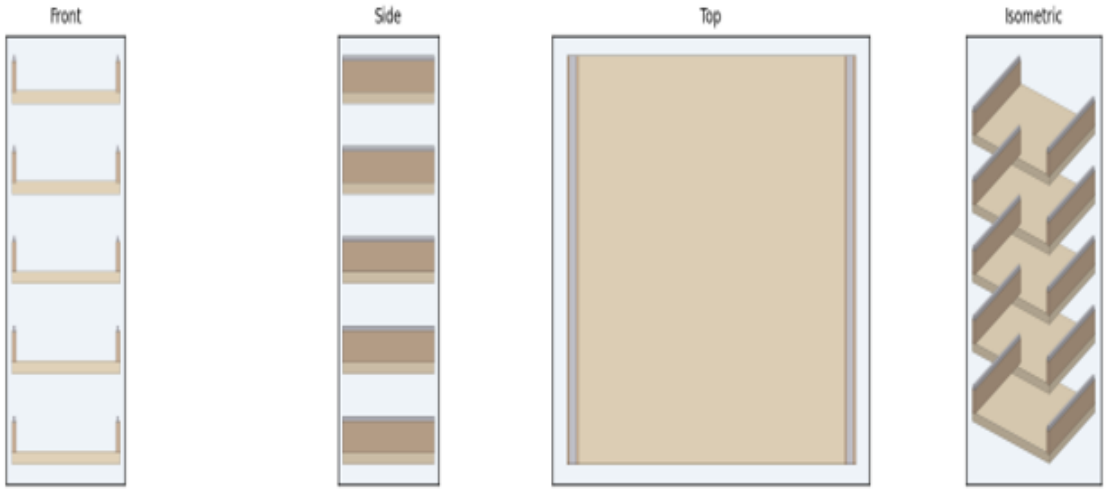


Figure 6: Interconnecting Corridors

6.4 Interconnecting Corridors

To unify the four independent blocks into a singular academic facility, three open-air skybridge corridors are inserted into the 2.0-unit gaps between adjacent blocks (z_1 and z_2):

- **Corridor Placement Math:** The midpoint along Z is calculated as $Z_{\text{mid}} = \frac{z_1 + z_2}{2.0}$, with a span length $\text{gapLength} = (z_2 - z_1) - 3.8 = 2.2$ units to fit flush between the block walls.

- **Multi-Story Floor Slabs (Floors 1 to 5):** For each floor level $f \in \{0, 1, 2, 3, 4\}$, a floor slab of width $X_{\text{width}} = 2.6$ is generated via translation $\mathbf{T}(x_{\text{offset}}, f \cdot 1.3, Z_{\text{mid}})$ and scale $\mathbf{S}(2.6, 0.18, \text{gapLength})$.

6.5 Front Extension

The front entrance of Block 4 ($Z = 9.0$) is designed with an architectural object:

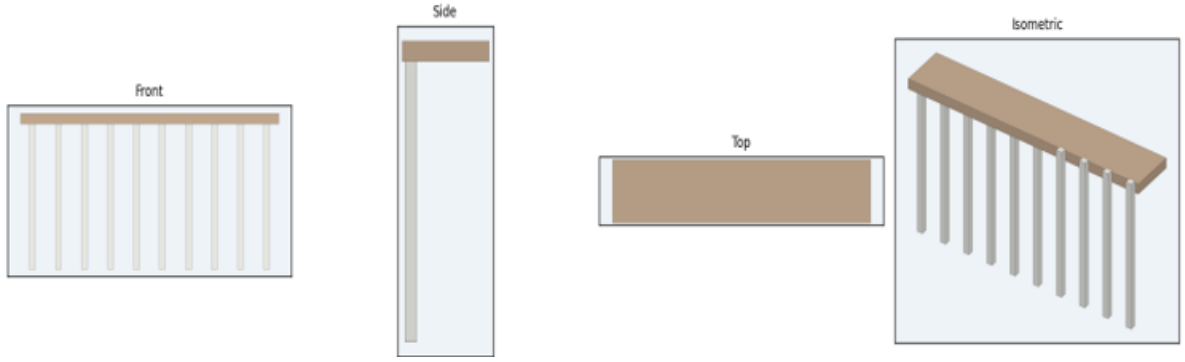


Figure 7: Front Extension

1. **Front Canopy:** A beige roof slab scaled to $\mathbf{S}(8.6, 0.25, 1.5)$ is positioned at height $Y = 3.6$ (spanning the lower two floors), extending from the front face $Z = 11.0$ to $Z = 12.5$.

2. **Pillars/ Columns:** Some white structural pillars of scale $\mathbf{S}(0.2, 3.6, 0.2)$ extend vertically from the ground ($Y = 0$) to the canopy ($Y = 3.6$) along the outer overhang edge ($Z = 12.35$). The X-coordinates are linearly interpolated across the front facade:

$$X_p = x_{\text{offset}} - 3.9 + \left(\frac{7.8}{9}\right) \cdot p \quad \text{for } p \in \{0, 1, \ldots, 9\}$$

3. **Facade Balcony Bands:** Horizontal accent ledges scaled to $\mathbf{S}(8.3, 0.18, 0.30)$ run across all 5 floors at $Z = 11.15$.

7 Architectural Modeling of Halls

7.1 AEH and SSH Halls

AEH (12×12, wall thickness 1.5) and SSH (12×15, wall thickness 1.8) are five-storey courtyard-type buildings: four wings surround an open yard. Walls are pink plaster $(0.88, 0.68, 0.72)$ with white concrete trim $(0.93, 0.93, 0.90)$ parapets, a dark plinth $(0.25, 0.22, 0.20)$, dark recessed windows $(0.18, 0.22, 0.28)$ and dark entrance doors $(0.12, 0.12, 0.14)$. Height is 5.83, divided into five storeys of 1.166. Window positions are computed from the wall length so they stay evenly distributed.

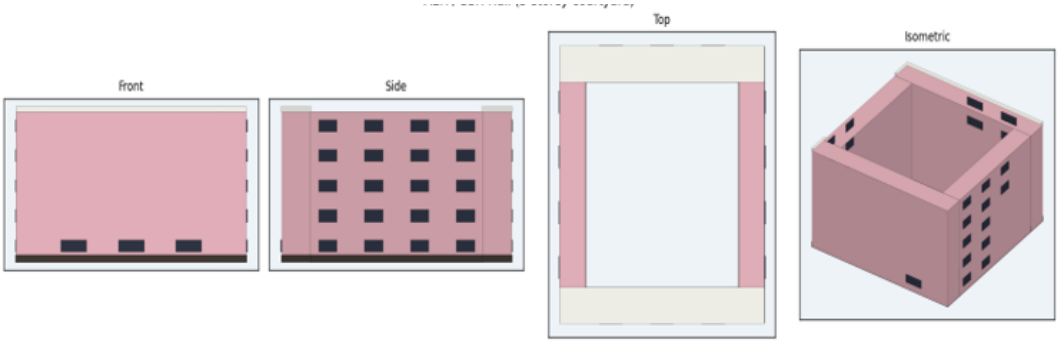


Figure 8: AEH / SSH hall: front, side, top and isometric views.

7.1.1 Night Lighting

When "hall window lights" are active (key `H`, night mode only), the windows change from dark glass to a warm orange $(1.0, 0.62, 0.26)$ and each hall adds four small point-type window lights (orange $1.0, 0.48, 0.19$) to the scene, giving a lit-window effect on nearby surfaces. The AEH/SSH window light can also be the shadow caster in this mode.

7.2 FFH, DRH and BRH Halls

FFH (at $-11, 3$) and DRH (at $-11, 23$) share the same drawing function: 7×10 footprint, height 3.75, wall thickness 1.2. Three wings are three storeys high ($h/3 = 1.25$ per storey) and the southern wing is only one storey, which creates a stepped silhouette and an open entry corner. BRH (at $27, 23$) uses the same layout rotated $180°$ about its centre, so its low wing faces the opposite direction.

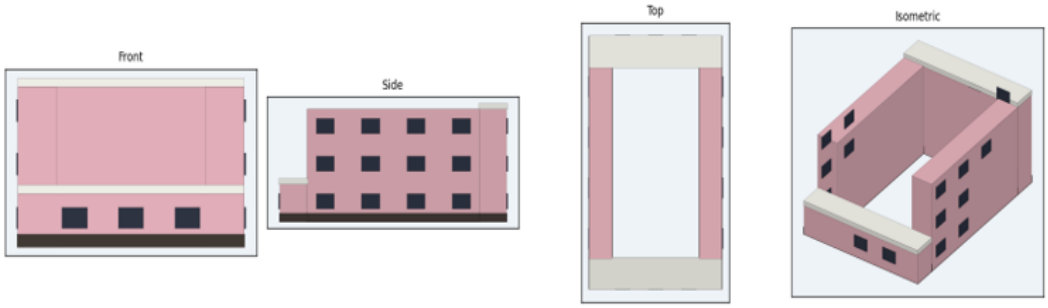


Figure 9: FFH / DRH / BRH hall: front, side, top and isometric views.

7.3 LHS Hall

The LHS hall (at $-13, -13$) is a long narrow building, 2 units wide and 10 units deep, height 3.75, with the same pink/white/dark palette, windows and trim as the other halls.

8 Vehicles

Both vehicles are constructed using a parent-child hierarchical matrix transformation tree. A root matrix defines the vehicle's position and orientation on the road network, and all

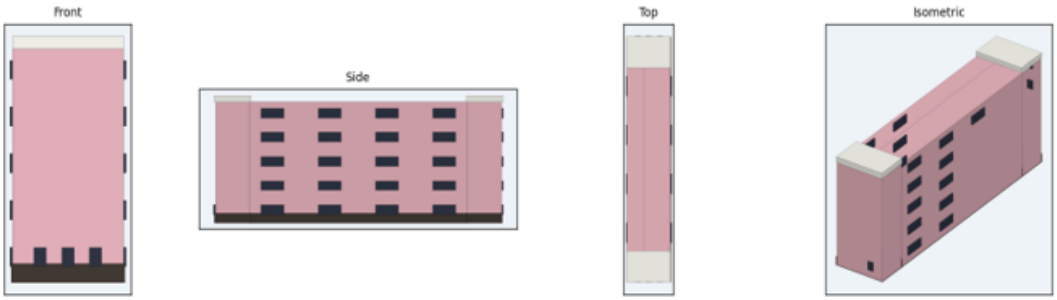


Figure 10: LHS hall: front, side, top and isometric views.

sub-components (chassis, roof, windshields, window panes, pillars, bumpers, headlights, tail lights, side mirrors, and wheels) inherit this root transformation.

8.1 Car Construction

1. **Root Matrix :** Translates to world position $(x, 0, z)$ and rotates by $0°$ (heading South $+Z$) or $180°$ (heading North $-Z$):

$$\mathbf{M}_{\text{car}} = \mathbf{T}(x, 0.0, z) \cdot \mathbf{R}_Y(\theta_{\text{heading}})$$

2. **Lower Chassis Shell:** Scaled to $\mathbf{S}(1.50, 0.45, 3.40)$ at local $\mathbf{T}(0.0, 0.42, 0.0)$.

3. **Bumpers:** Front bumper at local $\mathbf{T}(0.0, 0.30, +1.71)$ and rear bumper at local $\mathbf{T}(0.0, 0.30, -1.71)$ scaled to $\mathbf{S}(1.52, 0.22, 0.08)$.

4. **Cabin Roof & Support Frame:** Flat roof panel scaled to $\mathbf{S}(1.36, 0.05, 1.35)$ at $\mathbf{T}(0.0, 1.12, -0.15)$ resting on a cabin core frame $\mathbf{S}(1.34, 0.48, 1.35)$.

   - **Sloped Front & Rear Windshields:**
     - **Front Glass:** Sloped back by $-28°$ about the local X-axis:

       $$\mathbf{M}_{\text{windshield, front}} = \mathbf{M}_{\text{car}} \cdot \mathbf{T}(0.0, 0.90, 0.72) \cdot \mathbf{R}_X(-28.0°) \cdot \mathbf{S}(1.32, 0.52, 0.04)$$

     - **Rear Glass:** Sloped forward by $+26°$ about the local X-axis:

       $$\mathbf{M}_{\text{windshield, rear}} = \mathbf{M}_{\text{car}} \cdot \mathbf{T}(0.0, 0.90, -1.02) \cdot \mathbf{R}_X(+26.0°) \cdot \mathbf{S}(1.32, 0.50, 0.04)$$

5. **Side Windows & B-Pillars:** Two left windows $\mathbf{S}(0.03, 0.38, 0.55)$ at $X = -0.685$, two right windows at $X = +0.685$, separated by central B-pillars $\mathbf{S}(0.04, 0.40, 0.08)$ at $X = \pm 0.69$.

8.2 Bus Construction

1. **Root Matrix :** Establishes base frame $\mathbf{M}_{\text{bus}} = \mathbf{T}(x, 0.0, z) \cdot \mathbf{R}_Y(\theta_{\text{heading}})$.

2. **Lower Maroon Skirt:** Scaled to $\mathbf{S}(2.2, 0.7, 7.6)$ at local $\mathbf{T}(0.0, 0.55, 0.0)$.

3. **Horizontal Maroon Accent Stripe:** Continuous accent band scaled to $\mathbf{S}(2.22, 0.15, 7.62)$ at local $\mathbf{T}(0.0, 0.95, 0.0)$.

4. **Upper White Body & Roof Cap:** Upper cabin core scaled to $\mathbf{S}(2.18, 1.65, 7.58)$ at $\mathbf{T}(0.0, 1.85, 0.0)$ topped by a roof slab $\mathbf{S}(2.22, 0.12, 7.62)$ at $\mathbf{T}(0.0, 2.72, 0.0)$.

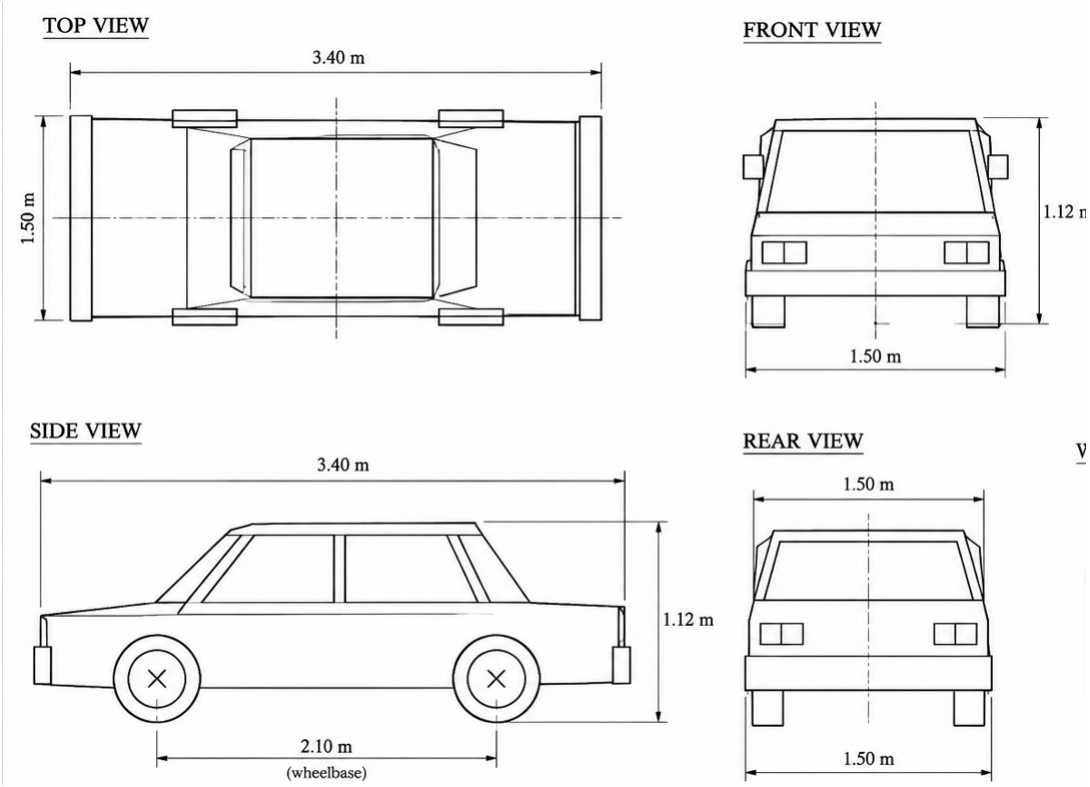


Figure 11: Car Architecture

8.3 3D Wheel Construction Mechanics

Each wheel is constructed as a 3D composite assembly of concentric cylinders and rotating cube spokes:

1. **Outer Rubber Tire:** Rendered with radius r and width w, colored dark matte rubber RGB $(0.10, 0.10, 0.10)$.

2. **Metallic Inner Rim:** A slightly wider, smaller-radius cylinder rendered inside the tire at $r_{\text{rim}} = 0.65 \cdot r$ and $w_{\text{rim}} = 1.02 \cdot w$ with bright silver color $(0.75, 0.75, 0.78)$.

3. **Rotating Cross-Spokes:** Two perpendicular thin unit cubes are scaled and multiplied by the wheel's local matrix:

$$\mathbf{M}_{\text{spoke 1}} = \mathbf{M}_{\text{wheel}} \cdot \mathbf{S}(1.05 \cdot w, 1.10 \cdot r, 0.18 \cdot r)$$

$$\mathbf{M}_{\text{spoke 2}} = \mathbf{M}_{\text{wheel}} \cdot \mathbf{S}(1.05 \cdot w, 0.18 \cdot r, 1.10 \cdot r)$$

Because spoke 1 and spoke 2 inherit $\mathbf{M}_{\text{wheel}}$, any pitch rotation $\mathbf{R}_X(\theta_{\text{spin}})$ applied to $\mathbf{M}_{\text{wheel}}$ automatically rotates the cross-spokes, creating a visible 3D wheel motion.

8.4 Motion Logic & Path Calculation

Vehicle updates and movement math occur inside C++ CPU loop functions.

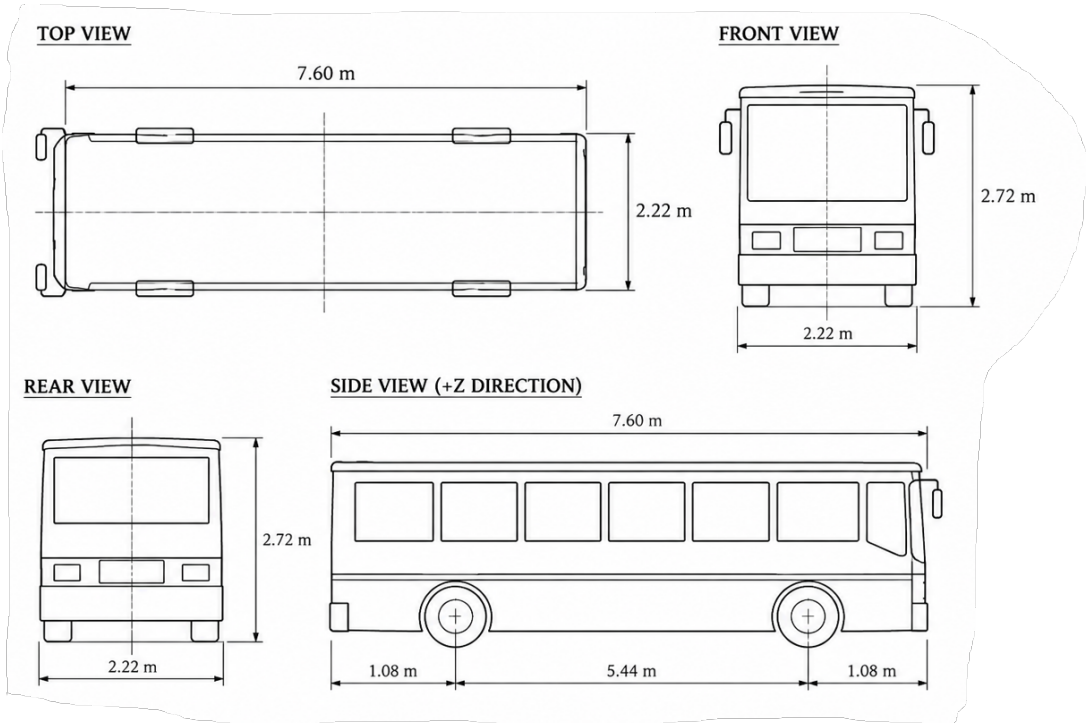


Figure 12: Bus Architecture

8.4.1 Continuous Highway Loop Arithmetic

Vehicles travel along the Main Central Road spanning $Z \in [-20.0, +46.0]$ (total road length $L = 66.0$ units) at speed $v = 5.0$ units/sec.

The travel position along a complete round trip ($2L = 132.0$ units) is evaluated using floating-point modulo arithmetic:

$$\text{cycle}(t) = (t \cdot v) \bmod (2L) = (t \cdot 5.0) \bmod 132.0$$

- **Phase 1 (cycle $<$ 66.0 — Southbound Journey):** Vehicle heading direction $d_z = +1.0$ (South towards viewer).
  - **World Z Position:** $Z(t) = -20.0 + \text{cycle}$.
  - **Lane Assignment:** $X = 5.25$ (West lane).
- **Phase 2 (cycle $\geq$ 66.0 — Northbound Journey):** Vehicle heading direction $d_z = -1.0$ (North away from viewer).
  - **World Z Position:** $Z(t) = -20.0+66.0-(\text{cycle}-66.0) = 46.0-(\text{cycle}-66.0)$.
  - **Lane Assignment:** $X = 1.75$ (East lane).

8.4.2 Two-Way Counter-Traffic Positioning

To simulate realistic two-way campus traffic without vehicle overlap, the Car and Bus run in opposite direction loops:

- **Bus Motion:** Position $(X_{\text{bus}}, Z_{\text{bus}})$ derived directly from `getAutomaticBusMotion(time)`.
- **Car Motion:** Assigned the inverted direction $-d_{z,\text{bus}}$, placing it in the opposite lane at position $Z_{\text{car}} = -20.0 + 66.0 - Z_{\text{bus}}$.

8.5 Wheel Rotation Physics & Mathematical Equations

Wheel rotation is linked to linear vehicle displacement under the pure rolling without slipping constraint.The linear distance s traversed by the vehicle equals the arc length rolled along the tire circumference:

$$s = v \cdot t = r \cdot \theta_{\text{rad}}$$

Where r is the effective wheel rolling radius and θ_{rad} is the accumulated angular rotation in radians. Converting θ to degrees for OpenGL matrix rotation calls ($\mathbf{R}_X(\theta_{\text{deg}})$) gives:

$$\theta_{\text{deg}} = \left(\frac{s}{r_{\text{eff}}}\right) \cdot \frac{180^\circ}{\pi} = \texttt{glm::degrees}\left(\frac{v \cdot t}{r_{\text{eff}}}\right)$$

9 Others

9.1 Durbar Bangla Side

The Durbar Banlga at $(15.2, 18)$ is not fully implemented. Just a shape has created. It constructed from three distinct geometric layers:

1. **Outer Base Slab:** A square ring slab ($8 \times 0.02 \times 8$) textured with seamless red-brick mapping (`seamless_brick.png`, UV scale 6×6).
2. **Inner Grass Median:** A raised square grass island ($4 \times 0.2 \times 4$) rendered in green $(0.2, 0.6, 0.2)$.
3. **Central Obelisk:** A concrete gray monument ($1 \times 2 \times 1$, color $(0.8, 0.8, 0.8)$) positioned at the center.

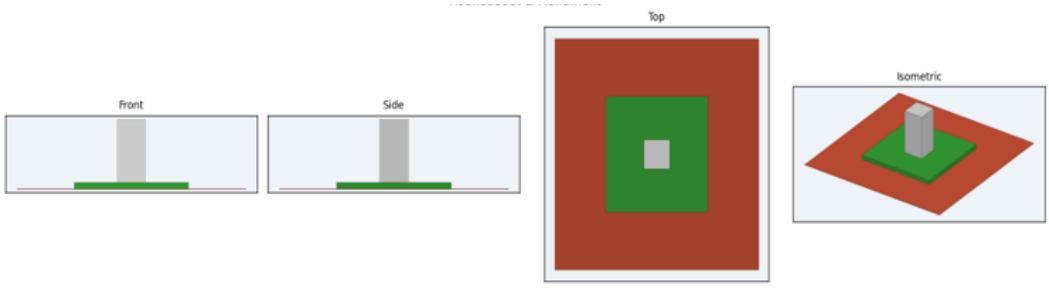


Figure 13: Durbar Bangla Side: front, side, top, and isometric views.

9.2 Trees

Each tree is modeled as a composite hierarchical mesh made of three primitives:

- **Trunk:** A brown cube ($0.4 \times 2.5 \times 0.4$, color $(0.35, 0.20, 0.10)$).
- **Lower Foliage Tier:** A green cube ($2 \times 1.5 \times 2$, color $(0.12, 0.50, 0.18)$).
- **Upper Foliage Tier:** A smaller green cube ($1.3 \times 1.2 \times 1.3$, color $(0.12, 0.50, 0.18)$) creating a stylized stepped crown.

A total of 23 trees are distributed across the campus: 8 along $X = -3.2$, 8 along $X = 8.8$, 5 along $X = 27.8$ (northern segment), and 2 adjacent to the roundabout at $(16, 22)$ and $(16, 31)$.

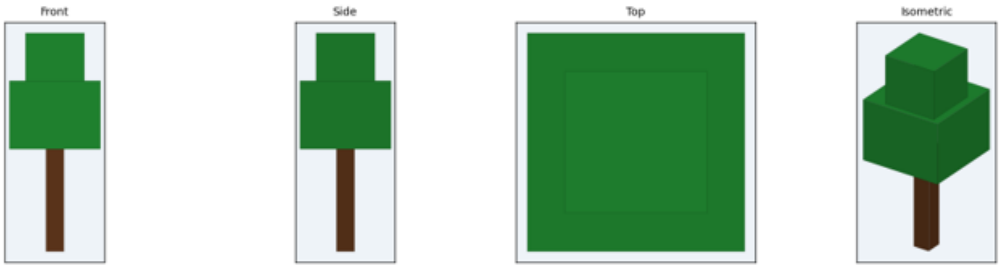


Figure 14: Tree model: front, side, top, and isometric views.

9.3 Street Lamp Posts

Each lamp post is a detailed composite structure consisting of a three-part gold pedestal, a dark cylindrical pole (radius 0.08, height 3.4 generated via `drawCylinder` rotated upright), a gold trim ring, a 2.2-unit cross-arm, a decorative crest and finial reaching a total height of 4.85 units, and two lamp heads mounted at ± 1 unit offset.

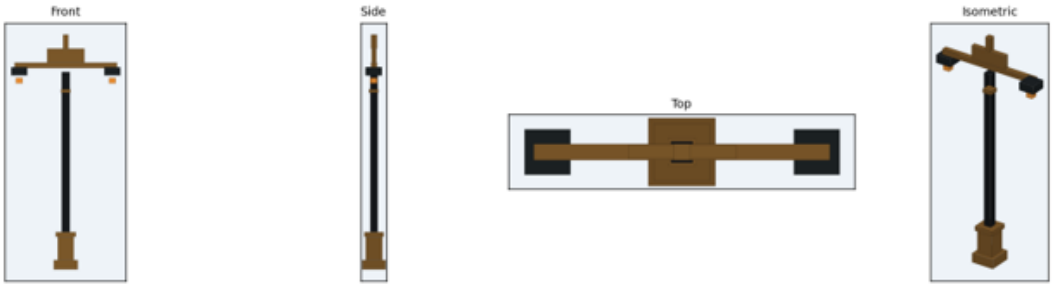


Figure 15: Street lamp post: front, side, top, and isometric views.

Each lamp head comprises:

- **Shade:** A dark ellipsoid shade (sphere primitive scaled to $0.34 \times 0.18 \times 0.34$).
- **Bulb:** A sphere primitive scaled to $0.14 \times 0.12 \times 0.14$. In night mode, lit bulbs use an emissive orange color $(1.0, 0.56, 0.20)$ with 0.88 opacity. When unlit, bulbs revert to gray $(0.24, 0.27, 0.29)$ at 0.58 opacity, rendered with alpha blending.

A total of 13 posts (26 lamp heads) are deployed across the campus: 10 along the main road at $X = -0.5$ and 7.5 ($Z \in \{-10, 2, 14, 22, 32\}$) and 3 along the back road at $X = 25.5$ ($Z \in \{-10, 2, 14\}$). Each lamp head serves as an individual downward-pointing spotlight source (26 spotlights total).

10 Lighting & Shading Calculations

10.1 Classification of Scene Light Sources

The lighting pipeline in supports up to 5 distinct light source categories, allowing full real-time control over scene illumination:

Lighting follows the Blinn–Phong model. For every light source, the function `illuminate()` returns $(I_{\text{diffuse}} + 0.25 \times I_{\text{specular}}) \times \mathbf{C}_{\text{light}} \times f_{\text{att}}$, where $I_{\text{diffuse}} = \max(\mathbf{N} \cdot \mathbf{L}, 0)$ and

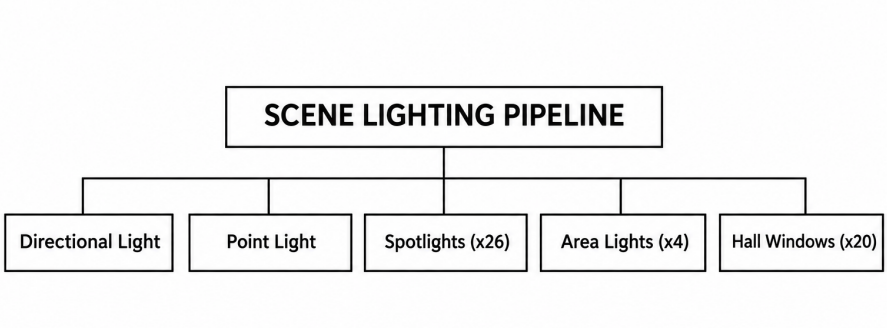


Figure 16: Scene Lighting Classification Hierarchy

$I_{\text{specular}} = (\max(\mathbf{N} \cdot \mathbf{H}, 0))^{32}$, with $\mathbf{H}$ being the half-vector between the light vector $\mathbf{L}$ and view direction $\mathbf{V}$. The final fragment color is given by $(k_a \cdot \mathbf{1} + \sum \mathbf{I}_{\text{light}}) \odot \mathbf{C}_{\text{base}}$.

Table 2: Summary of Scene Light Sources, Positions, Colors, and Attenuation Models

Light	**Position / direction**	**Colour**	**Attenuation / notes**
Table 2 continued from previous page			
Light	**Position / direction**	**Colour**	**Attenuation / notes**
Ambient	Global	White $\times 0.38$ (day) / 0.08 (night)	Constant base light
Directional (sun)	Direction = normalize$(-\mathbf{p}_{\text{sun}})$	$(1.0, 0.94, 0.82)$ day; $(0.12, 0.16, 0.30)$ night	No attenuation; toggle with ImGui
Point light	$(-10, 16, 2)$	$(0.55, 0.68, 1.0)$ bluish	$\frac{1.0}{1.0+0.02d+0.002d^2}$
Spot lights (26)	Lamp heads at $Y = 4.02$, pointing $(0, -1, 0)$	$(1.0, 0.78, 0.42)$ warm	Cone: inner cutoff $\cos 18°$, outer $\cos 28°$, smooth edge; attenuation $\frac{1.0}{1.0+0.04d+0.01d^2}$
Area light (4 sample points)	$(-18/-14, 18, 8/12)$	$(0.80, 0.42, 0.24)$	$\frac{0.25}{1.0+0.02d+0.002d^2}$ per sample — approximates a rectangular light above the halls
Hall window lights (20)	4 samples at each of 5 halls	$(1.0, 0.48, 0.19)$ orange	$\frac{0.05}{1.0+0.04d+0.008d^2}$; night mode only

The spot-light cone factor is evaluated as clamp $\left(\frac{\cos\theta - \text{outer}}{\text{inner} - \text{outer}}, 0.0, 1.0\right)$, where $\cos\theta$ is the cosine of the angle between the light direction vector and the vector from the light source to the fragment. This produces a smooth penumbra falloff. The area light is approximated

using four point samples placed at the corners of a 4×4 rectangle, providing an efficient real-time approximation of rectangular illumination.

1. **Global Directional Light (Sun / Moon):**

   - **Vector Definition:** Parallel light rays specified by direction vector $\mathbf{L}_{\text{dir}} = \text{normalize}(-\mathbf{p}_{\text{sun}})$.
   - **Attenuation:** Infinite range with zero distance attenuation .
   - **Environmental Point Light:**
     - **Position & Color:** Placed at global position $\mathbf{p}_{\text{point}} = (-10.0, 16.0, 2.0)$
     - **Distance Attenuation Model:** Quadratic decay based on fragment distance $d = |\mathbf{p}_{\text{point}} - \mathbf{p}_{\text{frag}}|$:

$$f_{\text{att, point}} = \frac{1.0}{1.0 + 0.02 \cdot d + 0.002 \cdot d^2}$$

   - **Street Lamp Spotlights :**
     - **Post Layout:** 26 spotlight heads mounted on 13 double-arm lamp posts at height $Y = 4.02$ across $X \in \{-0.5, 7.5, 25.5\}$ and $Z \in \{-10.0, 2.0, 14.0, 22.0, 32.0\}$.
     - **Light Direction:** Pointed vertically downward $\mathbf{D}_{\text{spot}} = (0.0, -1.0, 0.0)$
     - **Angular Cutoff Cone Clamping:**
       - Inner Cutoff Angle: $\theta_{\text{inner}} = 18^\circ$ ($\cos(18^\circ) \approx 0.951$).
       - Outer Cutoff Angle: $\theta_{\text{outer}} = 28^\circ$ ($\cos(28^\circ) \approx 0.883$).
       - Dual-Cone Falloff Factor (cone):

$$\text{cone} = \text{clamp}\left(\frac{\cos\theta - \cos\theta_{\text{outer}}}{\cos\theta_{\text{inner}} - \cos\theta_{\text{outer}}}, 0.0, 1.0\right)$$

       where $\cos\theta = \text{dot}(\text{normalize}(\mathbf{p}_{\text{frag}} - \mathbf{p}_{\text{spot}}), \mathbf{D}_{\text{spot}})$.
     - **Distance Attenuation & Cone Synthesis:**

$$f_{\text{att, spot}} = \frac{\text{cone}}{1.0 + 0.04 \cdot d + 0.01 \cdot d^2}$$

   - **Area Overhead Lights:**
     - **Quad Positions:** Arranged at roof level above Western halls $\mathbf{p}_{\text{area}} \in \{(-18, 18, 8), (-14, 18, 8), (-18, 18, 12), (-14, 18, 12)\}$ attenuation $\frac{0.25}{1.0+0.02d+0.002d^2}$.
   - **Hall Window Sampled Point Lights :**
     - **Positions & Color:** sampled emitters placed across building facades with steep localized attenuation $\frac{0.05}{1.0+0.04d+0.008d^2}$.

11 Shading Pipelines (Flat, Gouraud, Phong)

The rendering engine supports real-time switching between three classic graphics shading pathways via hotkeys (`F`, `G`, `P`) or the Dear ImGui control overlay:



(a) Flat Shading (b) Gouraud Shading (c) Phong Shading

Figure 17: Visual comparison of Flat, Gouraud, and Phong shading pipelines.

11.1 Flat Shading

- **Normal Vector Evaluation:** Surface normals are evaluated per fragment directly from geometry using screen-space partial derivative cross products:

$$\mathbf{N}_{\text{flat}} = \text{normalize}\left(\mathbf{dFdx}(\mathbf{p}_{\text{frag}}) \times \mathbf{dFdy}(\mathbf{p}_{\text{frag}})\right)$$

- **Visual Characteristics:** Discards smooth vertex normals, rendering every polygon face with a single uniform shade. Surfaces display sharp polygonal boundaries, giving a distinct low-poly faceted appearance.

11.2 Gouraud Shading

- **Vertex Shader Execution):** The full Blinn-Phong illumination model is calculated at every mesh vertex inside . Vertex shaders pass pre-computed ambient, directional, point, spotlight, area, and hall window lighting vectors down the pipeline.
- **Fragment Shader Interpolation :** Hardware rasterizers linearly interpolate color intensities across polygon interiors.
- **Performance & Artifacts:** Fast execution, but specular glints situated in the center of large low-tessellation polygons (e.g., ground plane slabs or broad building walls) are distorted or lost due to linear color interpolation between distant vertices.

11.3 Phong Shading

- **Vertex Shader Execution :** Vertex normals are transformed into world space via the inverse-transpose model matrix:

$$\mathbf{N}_{\text{world}} = \text{mat3}\left((\mathbf{M}^{-1})^{T}\right) \cdot \mathbf{aNormal}$$

- **Fragment Shader Execution :** Normals are re-normalized at every fragment . Light vectors **L**, view vectors **V**, halfway vectors **H**, cone cutoffs, and attenuation factors are computed independently for every single screen pixel.
- **Visual Characteristics:** Delivers smooth curved surfaces (e.g., sphere foliage, curved road junctions, vehicle windshields), exact spotlight cone boundaries, and accurate circular specular highlights across all scene geometry regardless of mesh density.

12 Texture

12.1 C++ Texture Class Architecture & stb_image Integration

The application implements an object-oriented texture management system through a custom C++ `Texture` class. This encapsulates image decoding, GPU memory allocation, texture unit binding, and resource cleanup:

(a) **Host-Side Image Decoding:** Image assets are read from disk using `stb_image.h`. The decoder executes `stbi_set_flip_vertically_on_load(true)` prior to parsing, flipping vertical pixel rows to align image origins with OpenGL's bottom-left UV coordinate space.

(b) **Directory Fallback Mechanism:** If an image fails to load from its primary path, the class executes an automatic fallback search in adjacent parent directories (`../image/`), preventing crashes caused by varying relative working directories.

(c) **Host Memory Cleanup:** Immediately after transferring raw pixel data to VRAM via `glTexImage2D`, the temporary CPU buffer is freed using `stbi_image_free` to optimize system RAM usage.

13 Camera and User Interaction

The projection is a perspective projection with a 45° field of view, near plane 0.1 and far plane 150, using the live window aspect ratio.

Table 3: Camera and User Interaction Controls

Input	Action
Tab / ImGui radio buttons	Switch between First-person and Orbit cameras.
`W A S D` (first person)	Walk forward / left / backward / right at 8 units per second on the ground plane; position clamped to $X \pm 35$, $Z \in [-35, 48]$. Eye height is 0.7, start at $(3.5, 0.7, 44)$.
Right mouse drag (first person)	Look around (0.12° per pixel, pitch limited to ±85°).
Left mouse drag (orbit)	Rotate around the scene (0.25° per pixel; pitch clamped 1°–89°; start radius 45, yaw 80°, pitch 10°).
Mouse wheel (orbit)	Zoom in and out, 3 units per notch, radius limited to 5–120.
`F` / `G` / `P`	Flat / Gouraud / Phong shading.
`L`	Toggle street lamp spot lights.
`H`	Toggle hall window lights (night mode only).
Up / Down arrow keys (manual bus)	Drive the bus when manual control is enabled.

Continued on next page

Table 3 continued from previous page

Input	Action
`Esc`	Exit the program.

The ImGui "Shading Controls" window offers the shading buttons, the camera radio buttons, manual bus control, the Day-mode checkbox, individual checkboxes for directional, point, spot and area lights, a sun-shadow checkbox, and read-outs of the current shadow caster, hall-window light state and active shading mode.

14 Conclusion

This project successfully developed an interactive 3D simulation of the KUET Academic Building and its surrounding campus using C++, OpenGL 3.3, GLSL, GLFW, and GLM. The scene includes academic buildings,coridoors, residential halls, roads, trees, streetlights, cars, and buses, all created using basic geometric shapes. Coordinate calculations were used to create symmetrical building designs. Different light sources, including sunlight, point lights, spotlights, and area lights, were implemented to create day and night scenes. Flat, Gouraud, and Phong shading modes, along with road and brick textures, were added to improve the visual quality. Real-time shadow mapping with PCF filtering was also implemented to produce smoother shadows. Additionally, orbit and first-person camera modes allow users to explore the campus interactively, while lighting, shading, and vehicle movement can be controlled. Overall, this project demonstrates the practical application of computer graphics techniques, including 3D modeling, transformations, lighting, shading, texturing, shadow mapping, and real-time rendering, to create an interactive virtual campus environment.